Q86) $f(x,y) = \sqrt{x^2+y^2}$

w.r.t x

$$f_x = \frac{2x}{2\sqrt{x^2+y^2}}$$

$$f_x = \frac{x}{x^2+y^2}$$

$$f_{xy} = x\,\partial/\partial y\,(x^2+y^2)^{-1/2}$$

$$= x\left(-\frac{1}{2}(x^2+y^2)^{-3/2}(2y)\right)$$

$$= -xy\,(x^2+y^2)^{-3/2}$$

w.r.t y

$$f_y = \frac{y}{\sqrt{x^2+y^2}}$$

$$f_{yx} = y\cdot\partial/\partial x\,(x^2+y^2)^{-1/2}$$

$$= y\left(-\frac{1}{2}(x^2+y^2)^{-3/2}\right)(2x)$$

$$= -xy\,(x^2+y^2)^{-3/2}$$

$f_{xy} = f_{yx} = -xy\,(x^2+y^2)^{-3/2}$ Hence proved

Q87) $f(x,y) = e^x \cos y$

Sol:- w.r.t x | w.r.t y

$f_x = e^x \cos y$ | $f_y = -e^x \sin y$

$f_{xy} = -e^x \sin y$ | $f_{yx} = -e^x \sin y$

$f_{xy} = f_{yx} = -e^{-x}\sin y$ Hence proved

Q89) $f_{x,y} = \ln(4x-5y)$

w.r.t x

$$f(x) = \frac{4}{4x-5y}$$

$$f_{xy} = 4\,\partial/\partial y\,(4x-5y)^{-1}$$

$$= 4(4x-5y)^{-2}\cdot 5$$

$$= \frac{20}{(4x-5y)^2}$$

w.r.t y

$$f_y = \frac{5}{4x-5y}$$

$$f_{yx} = 5\,\partial/\partial x\,(4x-5y)^{-1}$$

$$= 5(\cdot(4x-5y)^{-2})\cdot 4$$

$$= \frac{20}{(4x-5y)^2}$$

Hence proved

Show that function satisfies laplace's eq.

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

Q101) (a) $z = x^2 - y^2 + 2xy$

w.r.t x

$$\frac{\partial z}{\partial x} = 2x + 2y \Rightarrow \frac{\partial^2 z}{\partial x^2} = 2$$

w.r.t y

$$\frac{\partial z}{\partial y} = -2y + 2x \Rightarrow \frac{\partial^2 z}{\partial y^2} = (-2)$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 2 + (-2) = 0$$

True.

b) $z = e^x \sin y + e^y \cos x$

w.r.t x.

$$\frac{\partial z}{\partial x} = e^x \sin y - e^y \sin x$$

$$\frac{\partial^2 x}{\partial x^2} = e^x \sin y - e^y \cos x$$

w.r.t y

$$\frac{\partial z}{\partial y} = e^x \cos y + e^y \cos x$$

$$\frac{\partial^2 x}{\partial y^2} = -e^x \sin y + e^y \cos x$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^x \cancel{\cos}\,\overset{\sin}{} y - \cancel{e^y \cos x} + \cancel{e^y \cos x} - \cancel{e^x \sin y}$$

$= 0$ True.

(c) $z = \ln(x^2+y^2) + 2\tan^{-1}(y/x)$

Sol:-

w.r.t x

$$\frac{\partial z}{\partial x} = \frac{2x}{x^2+y^2} + \frac{2}{1+y^2/x^2}\left(-y/x^2\right)$$

$$= \frac{2x}{x^2+y^2} - \frac{2\cancel{x^2}}{x^2+y^2}\,\cancel{x^2}\,y$$

$$= \frac{2x-2y}{x^2+y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{(x^2+y^2)\,\partial/\partial x\,(2x-2y) - 2x-2y)\,\partial/\partial x\,(x^2+y^2)}{(x^2+y^2)}$$

$$= \frac{2(\,(x^2+y^2) - 2x^2 + 2xy))}{x^2+y^2}$$

$$= \frac{2(y^2 - x^2 + 2xy)}{x^2+y^2}$$

$$= \frac{-2(x^2 - y^2 - 2xy)}{(x^2+y^2)}$$

z w.r.t y

$$\partial x = \frac{2y}{x^2+y^2} + \frac{2}{x^2+y^2} \cdot \frac{1}{x}$$

$$= \frac{2y + 2x}{x^2+y^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{(x^2+y^2)\,\partial/\partial y\,(2y+2x) - (2y+2x)\,\partial/\partial y\,(x^2+y^2)}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)(2) - (2)(y+x)(2y)}{(x^2+y^2)^2}$$

$$= \frac{-2(2y^2 + 2x^2 - 2x^2 - 2y^2)}{(x^2+y^2)^2}$$

$$= \frac{-2(y^2 - 2xy - x^2)}{(x^2+y^2)^2}$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = -2\left(\frac{-x^2+y^2-2xy+2xy-y^2+x^2}{(x^2+y^2)^2}\right)$$

$= 0$ True.

13.5

Chain rule

If $x = x(t)$ & $y = y(t)$ are differentiable at t & if $z = f(x, y)$ at point $(x, y) = (x(t), y(t))$

The $z = f(x(t), y(t))$ is differentiable at t &

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{\partial t} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Example 1

$z = x^2 y$, $x = t^2$, $y = t^3$ find $\frac{dz}{dt}$ by chain rule

formula = ~~find~~ $\frac{dz}{dt} \frac{dx}{dt} + \frac{dz}{dy} \frac{dy}{dt}$

$$\frac{dz}{dt} = (2xy)(2t) + (x^2)(3t^2)$$

$$= (2t^2 t^3)(2t) + (t^4)(3t^2)$$

$$= 4t^6 + 3t^6$$

$$= 7t^6$$

Example 2

$$w = \sqrt{x^2+y^2+z^2}$$

$w = \cos\theta$, $y = \sin\theta$, $z = \tan\theta$ when $\theta = \frac{\pi}{4}$

find $\frac{\partial w}{\partial \theta}$ by chain rule

Sol:-

$$\frac{dw}{\partial\theta} = \frac{\partial w}{\partial x}\frac{dx}{d\theta} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial\theta} + \frac{\partial w}{dz}\frac{dz}{d\theta}$$

$$= \sqrt{-1} \sin\theta$$

$$= \frac{1}{2}(x^2+y^2+z^2)^{-1/2}(2x)(-\sin\theta) + \frac{1}{2}$$

$$(x^2+y^2+z^2)^{-1/2}(2y)(\cos\theta)$$

$$+\frac{1}{2}(x^2+y^2+z^2)^{-1/2}(2z)(\sec^2\theta)$$

$$\theta = \frac{\pi}{4}$$

$$x = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}} , \quad y = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$z = \tan\frac{\pi}{4} = 1$$

$$\left.\frac{\partial w}{\partial \theta}\right|_{\theta=\pi/4} = \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)\left(\sqrt{2}\right)\left(\frac{-1}{\sqrt{2}}\right) + \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)(\sqrt{2}$$

$$+ \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)(2)(2)$$

$$= \sqrt{2} \quad \text{Ans}$$

when $\theta \neq \pi/4$

$$= -\frac{1}{2}\left(\cos^2\theta + \sin^2\theta + \tan^2\theta\right)^{-1/2}(2\cos\theta)(\sin\theta)$$

$$+ \frac{1}{2}\left(\cos^2\theta + \sin^2\theta + \tan^2\theta\right)^{-1/2}(2\sin\theta)(\cos\theta) +$$

$$\frac{1}{2}\left(\cos^2\theta + \sin^2\theta + \tan^2\theta\right)^{-1/2}(2\tan\theta)(\sec^2\theta)$$

Example 3

$z = e^{xy}$, $x = 2u+v$, $y = \frac{u}{v}$

find $\frac{dz}{du}$ & $\frac{dz}{dv}$ by chain rule

Sol:- $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial u} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$

$$\frac{\partial z}{\partial u} = ye^{xy}(2) + (xe^{xy})\left(\frac{1}{v}\right)$$

$$= \left[2y + \frac{x}{v}\right] e^{xy}$$

$$= \left[\frac{2u}{v} + \frac{2u+v}{v}\right] e^{(2u+v)(u/v)}$$

$$= \left[\frac{4u}{v} + 1\right] e^{(2u+v)(u/v)}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$= (ye^{xy})(1) + (xe^{xy})\left(-\frac{u}{v^2}\right)$$

$$= \left[y - x\left(\frac{u}{v^2}\right)e^{xy}\right] e^{xy}$$

$$= \left[\frac{u}{v} - (2u+v)\left(\frac{u}{v^2}\right)\right] e^{(2u+v)(u/v)}$$

$$= -\frac{2u^2}{v^2} e^{(2u+v)(u/v)}$$

Example 4

$w = e^{xyz}$, $x = 3u+v$

$y = 3u-v$

$z = u^2 v$

Find $\frac{\partial w}{\partial u}$, $\frac{\partial w}{\partial v}$

Sol:-

$$\frac{\partial w}{\partial u} = \frac{\partial w}{dx} \cdot \frac{dx}{\partial u} + \frac{\partial w}{dy} \frac{dy}{\partial u} + \frac{dw}{\partial z} \frac{dz}{\partial u}$$

$$= e^{xyz}(1)(yz)(3) + e^{xyz}(1)(xz)(3) + e^{xyz}(1)xy \, 2u(v)$$

$$= e^{xyz}$$

$$= 3yze^{xyz} + 3xze^{xyz} + 2uve^{xyz}xy$$

$$= e^{xyz}(3yz + 3xz + 2uvxy)$$

$$= e^{(3u+v)(3u-v)(u^2v)}\left(3(3u-v)(u^2v) + 3(3u+v)(u^2v) + 2uv(3u+v)(3u-v)\right)$$

Ans

$$= e^{(3u+v)(3u-v)(u^2v)} \left[(9u-3v)(3u^2v) + (9u+3v)(3u^2v) + 6u^2v + 2uv^2)(6u^2v - 2u^2) \right]$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{dx}{dv} + \frac{\partial w}{\partial y}\frac{\partial y}{dv} + \frac{\partial w}{\partial z}\frac{\partial z}{dv}$$

$$= e^{xyz}(yz)(1) + e^{xyz}(xz)(-1) + e^{xyz}(xy)u^2$$

$$= e^{xyz}yz - e^{xyz}xz + xye^{xyz}u^2$$

$$= e^{(3u+v)(3u-v)(u^2v)} \left[(3u-v)(u^2v) + (3u+v)(u^2v) + (3u+v)(3u-v)(u^2v) \right]$$

Example 5

$$w = x^2 + y^2 - z^2 \qquad x = \rho \sin\phi \cos\theta$$
$$y = \rho \sin\phi \sin\theta$$
$$z = \rho \cos\phi$$

find $\frac{\partial w}{\partial \rho}$ & $\frac{\partial w}{\partial \theta}$

sol:-

$$\frac{\partial w}{\partial \rho} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \rho} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \rho} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial \rho}$$

$$= 2x(\sin\phi\cos\theta) + 2y(\sin\phi\sin\theta) + (-2z)(\cos\phi)$$

$$= 2(\rho\sin\phi\cos\theta)(\sin\phi\cos\theta) + 2(\rho\sin\phi\sin\theta)(\sin\phi\sin\theta) - 2(\rho\cos\theta)(\cos\phi)$$

$$= 2\rho\sin^2\phi\cos^2\theta + 2\rho\sin^2\phi\sin^2\theta + -2\rho\cos^2\phi$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial \theta}$$

$$= -(2x)(\rho\sin\phi\sin\theta) + (2y)(\rho\sin\phi\cos\theta) - (2z)(0)$$

$$= -2x\rho\sin\phi\sin\theta + 2y\rho\sin\phi\cos\theta$$

$$= 0$$

Example 6

$w = xy + yz \qquad y = \sin x$
$\qquad\qquad\qquad\qquad z = e^x$

~~So~~ Find $\frac{\partial w}{\partial x}$

Sol: $\frac{\partial w}{\partial x}\frac{\partial x}{\partial x}$

$$\frac{\partial w}{\partial x} = \left[\frac{\partial w}{\partial y}\frac{dy}{dx} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x}\right.$$

$$= y + (x+z)\cos x + y e^x$$

$$= y + (x + e^x)\cos x + \sin x \, e^x$$

Example 7

$$x^3 + y^2 x - 3 = 0$$

find $\frac{dy}{dx}$

Sol:-

$$\frac{\partial y}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial y} = -\frac{3x^2+y^2}{2xy}$$

Alternatively,

$$3x^2+y^2+x\left(2y\frac{dy}{dx}\right)-0=0$$

$$\frac{dy}{dx}=-\frac{3x^2+y^2}{2yx}$$

Theorems

$$\frac{\partial z}{\partial x}=-\frac{\partial f/\partial x}{\partial f/\partial z}$$

$$\frac{\partial z}{\partial y}=-\frac{\partial f/\partial y}{\partial f/\partial z}$$

Example 8

consider the sphere $x^2+y^2+z^2=1$ $\partial z/\partial x$ & $\partial z/\partial y$ at the point $\left(\frac{2}{3},\frac{1}{3},\frac{2}{3}\right)$

sol:

$$\frac{\partial z}{\partial x}=\frac{\partial f/\partial x}{\partial f/\partial z}=-\frac{2x}{2z}=-\frac{x}{z}$$

$$\frac{\partial z}{\partial y}=-\frac{\partial f/\partial y}{\partial f/\partial z}=-\frac{2y}{2z}=\frac{-y}{z}$$

At the point $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$ evaluating these derivatives gives

$$\frac{\partial z}{\partial x} = \frac{-x}{z} = -\frac{2/3}{2/3} = \boxed{-1}$$

$$\frac{\partial z}{\partial y} = \frac{-y}{z} = -\frac{1/3}{2/3} = \boxed{-\frac{1}{2}}$$

Q: Use an appropriate form of the chain rule to find $\frac{dz}{dt}$

1. $z = 3x^2y^3$; $x = t^4$, $y = t^2$

Sol:-

$$\frac{dz}{dt} = \frac{dz}{\partial x}\frac{dx}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= (6xy^3)(4t^3) + (9x^2y^2)(2t)$$

$$= (6t^4t^6)(4t^3) + 9(t^8t^4)(2t)$$

$$= 24t^{13} + 18t^{13}$$

$$= 42t^{13}$$

Q2) $z = \ln(2x^2+y)$

$x = \sqrt{t}$, $y = t^{2/3}$

Sol:-

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{1}{(2x^2+y)}(4x)\left(\frac{1}{2}t^{-1/2}\right) + \left(\frac{1}{2x^2+y}\right)\left(\frac{2}{3}t^{-1/3}\right)$$

$$= \frac{2x\,t^{-1/2}}{2x^2+y} + \frac{2}{3}\left(\frac{t^{-1/3}}{2x^2+y}\right)$$

$$= \frac{2\sqrt{t}\,t^{-1/2}}{2(t)+t^{2/3}} + \frac{2}{3}\left(\frac{t^{-1/3}}{2(t)+t^{2/3}}\right)$$

$$= \frac{2}{2t+t^{2/3}} + \frac{2}{3}\left(\frac{t^{-1/3}}{2t+t^{2/3}}\right)$$

$$= \frac{2}{2t+t^{2/3}}\left(1 + \frac{t^{-1/3}}{3}\right)$$

$$= \frac{2}{2t+t^{2/3}}\left(\frac{3+t^{-1/3}}{3}\right)$$

$$= \frac{2}{3}\left(\frac{3+t^{-1/3}}{2t+t^{2/3}}\right)$$

Q3) $z = 3\cos x - \sin xy$ $x = \frac{1}{t}$, $y = 3t$

Sol:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \left(\left(-3\sin x - \cos xy\right)(y)\left(-t^{-2}\right)\right) + \left(-\cos x^2 y\right)(3)$$

$$= (3\sin x + \cos xy^2)(t^{-2}) - 3\cos x^2 y$$

$$= 3t^{-2}\sin x + t^{-2}\cos xy^2 - 3\cos x^2 y$$

$$= 3t^{-2}\sin(1/t) + t^{-2}\cos(1/t)(9t^2) - 3\cos\left(\frac{1}{t^2}\right)(3t)$$

$$= 3t^{-2}\sin(1/t) + \cancel{9\cos\left(\frac{1}{t}\right) - 9\cos\left(\frac{1}{t}\right)}$$

$$= 3t^{-2}\sin(1/t)$$

Q4) $z = \sqrt{1 + x - 2xy^4}$, $x = \ln t$, $y = t$

Sol:-

$$\frac{dz}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{1}{2\sqrt{1 + x - 2xy^4}}(1 - 2y^4)\left(\frac{1}{t}\right) +$$

$$\frac{1}{2\sqrt{1 + x - 2xy^4}}(-2)(4y^3)$$

$$= \frac{1-2y^4}{2t\sqrt{1+x-2xy^4}} - \frac{8xy^3}{2\sqrt{1+x-2xy^4}}$$

$$= \frac{1}{2\sqrt{1+x-2xy^4}}\left(\frac{1-2y^4}{t} - 8xy^3\right)$$

$$= \frac{1}{2\sqrt{1+x-2xy^4}}\left(\frac{1-2y^4-8xy^3t}{t}\right)$$

$$= \frac{1-2t^4-8t^4\ln t}{2\sqrt{1+\ln t-2t^4\ln t}} \quad \text{Ans.}$$

(Q7)-(20) Use appropriate form of chain rule to find dw/dt

Q7) $w = 5x^2y^3z^4 \quad x=t^2, y=t^3, z=t^5$

Sol:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= (10xy^3z^4)(2t) + (15x^2y^2z^4)(3t^2) + (20x^2y^3z^3)(5t^4)$$

$$= (10t^2t^9t^{20})(2t) + (15t^4t^6t^{20})(3t^2) + (20t^4t^9t^{15})(5t^4)$$

$$= 20t^{32} + 45t^{32} + 100t^{32} = \boxed{165\ t^{32}}$$

$\frac{2}{3}-1 \quad \frac{2-3}{3} = \frac{-1}{3}$

Q8) $w = \ln(3x^2 - 2y + 4z^3) \quad x = t^{1/2}, y = t^{2/3}; z = t^{-2}$

Sol:-

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= \frac{6x}{(3x^2-2y+4z^3)}\frac{1}{2\sqrt{t}} - \frac{2}{3x^2-2y+4z^3}\frac{2}{3}t^{-1/3}$$

$$- \frac{12z^2}{3x^2-2y+4z^3}\, 2t^{-3}$$

$$= \left(\frac{6\sqrt{t}}{3t-2t^{2/3}+4t^{-6}}\right)\left(\frac{1}{2\sqrt{t}}\right) - \frac{4t^{-1/3}}{3(3t-2t^{2/3}+4t^{-6})}$$

$$- \frac{12t^{-4}}{3t-2t^{2/3}+4t^{-6}}(2t^{-3})$$

$$= \frac{1}{3t-2t^{2/3}+4t^{-6}}\left(3 - \frac{4t^{-1/3}}{3} - 24t^{-7}\right)$$

Q9) $w = 5\cos xy - \sin xz \qquad x = \frac{1}{t}, y = t, z = t^3$

Sol:-

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= \left[-5y\sin xy - z\cos xz\,(-t^{-2})\right] - 5x\sin xy(1)$$

$$- x\cos xz\,(3t^2)$$

$$= (5y\sin xy + z\cos xz)(t^{-2}) - 5x\sin xy - x\cos xz(3t^2)$$

$$= \left[5t\sin(1/t)(t) + t^3\cos(1/t)(t^3)\right](t^{-2})$$

$$-5(1/t)\sin(1/t)^{(t)} - (1/t)\cos(1/t)(t^3)(3t^2)$$

$$= 5t^{-1}\sin(1) + t\cos t^2 - 5(1/t)\sin(1) - 3t\cos t^2$$

$$= -2t\cos t^2 \quad \text{Ans}$$

Q10) $w = \sqrt{1+x-2yz^4x}$ $\quad x = \ln t$

$y = t, z = 4t$

Sol:-

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t}$$

$$= \frac{1-2yz^4}{2\sqrt{1+x-2yz^4x}}\left(\frac{1}{t}\right) + \frac{-2z^4x}{2\sqrt{1+x-2yz^4x}} + \left(\frac{-8yz^3x}{2\sqrt{1+x-2yz^4x}}\right)(4)$$

$$= \frac{1}{2\sqrt{1+x-2yz^4x}}\left(\frac{1-2yz^4}{t} - 2xz^4 - 32yz^3x\right)$$

$$= \frac{1}{2\sqrt{1+\ln t-2(t)(4t)^4\ln t}}\left(\frac{1-2t(4t)^4 - 2\ln t(4t)^4 - 32(t)(4t^3)(\ln t)}{t}\right)$$

$$= \frac{1}{2\sqrt{1+\ln t - 512t^5\ln t}}\left(\frac{1-512t^5-512t^5\ln t - 2048t^5\ln t}{t}\right)$$

$$= \frac{1-512t^5-2560t^5\ln t}{2t\sqrt{1+\ln t-512t^5\ln t}}$$

Use appropriate forms of chain rule to find derivative of.

Q27) $z = \ln(x^2+1)$ find $\frac{\partial z}{\partial r}$ & $\frac{\partial z}{\partial \theta}$

$x = r\cos\theta$

Sol:-

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\,\frac{\partial x}{\partial r}$$

$$= \frac{1}{x^2+1}(2x)(\cos\theta)$$

$$= \frac{2x\cos\theta}{x^2+1} \Rightarrow = \frac{2r\cos^2\theta}{r^2\cos^2\theta+1}$$

$$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\,\frac{\partial x}{\partial \theta}$$

$$= \frac{2x}{x^2+1}(-r\sin\theta)$$

$$= \frac{-2xr\sin\theta}{x^2+1}$$

$$= \frac{-2r^2\cos\theta\sin\theta}{r^2\cos^2\theta+1}$$

Q28) $u = rs^2 \ln t$ Find $\frac{\partial u}{\partial x}$ & $\frac{\partial x}{\partial y}$

$r = x^2$

$s = 4y+1$

$t = xy^3$

Sol:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial x}$$

$$= (s^2 \ln t)(2x) + \left(\frac{rs^2}{t}\right) y^3$$

$$= 2xs^2 \ln t + \frac{rs^2 y^3}{t}$$

$$= 2x\,(4y+1)^2 \ln(xy^3) + \frac{x^2(4y+1)^2 y^3}{xy^3}$$

$$= \frac{2x^2y^3(4y+1)^2 \ln(xy^3) + x^2(4y+1)^2 y^3}{xy^3}$$

$$= x^2y^3(4y+1)^2\left(\frac{2\ln(xy^3)+1}{xy^3}\right)$$

$$= x\,(4y+1)^2\left(1+2\ln(xy^3)\right)$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial u}{\partial t}\frac{\partial t}{\partial y}$$

$$= (2rs \ln t)(4) + \left(\frac{rs^2}{t}\right)(3xy^2)$$

$= 8rs\ln t + \frac{3xy^2rs^2}{t}$

$= 8x^2(4y+1)(\ln t)\cancel{(4)} + \frac{3xy^2x^2(4y+1)^2}{xy^3}$

$= 8x^2\ln(xy^3)(4y+1) + \frac{3x^3y^2(4y+1)^2}{xy^3}$

$= 8x^2(4y+1)\ln xy^3 + \frac{3x^2(4y+1)^2}{y}$

Q29) $w = 4x^2 + 4y^2 + z^2$, $x = \rho\sin\phi\cos\theta$

$y = \rho\sin\phi\sin\theta$

$z = \rho\cos\phi$

Find $\frac{\partial w}{\partial \rho}$, $\frac{\partial w}{\partial \phi}$ & $\frac{\partial w}{\partial \theta}$

Sol:-

$\frac{dw}{\partial \rho} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \rho} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \rho} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \rho}$

$= 8x(\sin\phi\cos\theta) + 8y(\sin\phi\sin\theta) + 2z(\cos\phi)$

$= 8\rho(\sin^2\phi\cos^2\theta) + 8\rho(\sin^2\phi\sin^2\theta) + 2\rho(\cos^2\phi)$

$= 2\rho(4\sin^2\phi\cos^2\theta + 4\sin^2\phi\sin^2\theta + \cos^2\phi)$

$= 2\rho(4\sin^2\phi(\sin^2\theta + \cos^2\theta) + \cos^2\phi)$

$= 2\rho(4\sin^2\phi + \cos^2\phi)$

$$\frac{\partial w}{\partial \phi} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \phi} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \phi} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \theta}$$

$$= 8x\,(\rho\cos\phi\cos\theta) + 8y\,(\rho\cos\phi\sin\theta) + 2z(\rho\sin\phi$$

$$= 8\rho^2(\sin\phi\cos\phi\cos^2\theta) + 8\rho^2(\sin\phi\cos\phi\sin^2\theta) - 2\rho^2(\sin\phi\cos\phi)$$

$$= -2\rho^2(4\sin\phi\cos\phi\cos^2\theta + 4\sin\phi\cos\phi\sin^2\theta - \sin\phi\cos\phi)$$

$$= -2\rho^2(4\sin\phi\cos\phi - \sin\phi\cos\phi)$$

$$= 2\rho^2(\quad -3\sin\phi\cos\phi)$$

$$= 6\rho^2 \sin\phi\cos\phi$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial \theta}$$

$$= 8x(-\rho\sin\phi\sin\theta) + 8y(\rho\sin\phi\cos\theta) + \cancel{\partial z}$$

$$= -8\rho^2(\sin^2\phi\cos\theta\sin\theta) + 8\rho^2(\sin^2\phi\cos\theta\sin\theta)$$

$$= 0 \quad \text{Ans.}$$

Find $\partial z/\partial x$ & $\partial z/\partial y$ by diff. and confirm that results obtained agree with those predicted by formulas in theorem.

Theorem -- $\dfrac{\partial f/\partial x}{\partial f/\partial z} = \dfrac{\partial z}{\partial x}$

Q46) $\ln(1+z) + xy^2 + z - 1 = 0$

Sol:-

$$\frac{\partial z}{\partial x} = \frac{-\partial f/\partial x}{\partial f/\partial z}$$

$$= -\frac{y^2}{\frac{1}{1+z}+1} \Rightarrow = -\frac{y^2}{\frac{1+1+z}{1+z}}$$

$$= -\frac{y^2(1+z)}{2+z}.$$

$$\frac{\partial z}{\partial y} = \frac{-\partial f/\partial y}{-\partial f/\partial z}$$

$$= -\frac{2xy}{\frac{1}{1+z}+1} = \frac{-2xy}{\frac{1+1+z}{1+z}}$$

$$= \frac{-2xy(1+z)}{2+z}$$

Q47) $ye^x - 5\sin 3z = 3z$

Sol:

$$= ye^x - 5\sin 3z - 3z$$

$$\frac{\partial z}{\partial x} = \frac{-\partial f/\partial x}{\partial f/\partial z}$$

$$= \frac{-ye^x}{-5\cos 3z(3)-3}$$

$$= \frac{ye^x}{15\cos 3z+3}$$

$$\frac{\partial z}{\partial y} = \frac{-\partial f/\partial y}{-\partial f/\partial z}$$

$$= \frac{-e^x}{15\cos 3z+3}$$

Q48) $e^{xy}\cos yz - e^{yz}\sin xz + 2$

Sol:-

$$\frac{\partial z}{\partial x} = \frac{-\partial f/\partial x}{\partial f/\partial z}$$

$$= -\left(\frac{e^{xy}(y)\cos yz - e^{yz}\cos xz(z)}{-e^{xy}\sin yz(y) - e^{yz}\cos xz(x) + \sin xz(e^{yz})(y)}\right)$$

$$= \frac{-ye^{xy}\cos yz + ze^{yz}\cos xz}{ye^{xy}\sin yz + xe^{yz}\cos xz + ye^{yz}\sin xz}$$

$$\frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z}$$

$$= -\left(\frac{e^{xy}(-\sin yz)(z) + \cos yz\,(e^{xy})(x) - e^{yz}(\sin xz)(z)}{-ye^{xy}\sin yz - xe^{yz}\cos xz - ye^{yz}\sin xz}\right)$$

$$= \frac{-ze^{xy}\sin yz - xe^{xy}\cos yz + ze^{yz}\sin xz}{ye^{xy}\sin yz + xe^{yz}\cos xz + ye^{yz}\sin xz}$$